

P1 — Consolidated Results

Scope

The final analytical sample contains **512 subnational regions in 28 countries**, observed annually from **2015 to 2019**. The additional candidate region outside the complete five-year analytical sample is not part of Scripts 04–07 final results.

Global M-Exp-FCMd solution

- Selected number of clusters: **C = 2**.
- Selected fuzziness parameter: **m = 2.5**.
- Fuzzy Silhouette: **0.699295**.
- Hard cluster sizes: **C1 = 100, C2 = 412**.
- Robust exponential beta: **205.415040**.
- Clustering uses only the three final dynamic family-level features; structural/static variables remain contextual and are not part of the clustering distance.

Dynamic clustering features

- `bod_demographic_productive_potential`
- `bod_socioeconomic_deprivation`
- `ntl_per_capita_norm`

Stability

The global bootstrap completed **500/500** successful replications. The mean hard-partition Adjusted Rand Index was **0.566100**, while mean fuzzy-membership similarity was **0.872783**. These statistics describe different aspects of stability and are not used to re-select *C*.

Fuzzy transition

83 regions (16.21%) have a membership margin below 0.20 and are therefore flagged as fuzzy-transition regions.

Temporal convergence

182 regions moved relatively closer to the alternative fixed global medoid profile between 2015 and 2019. This is a directional convergence measure and **does not represent formal cluster reassignment**.

- **C1->C2**: 37/100 regions (37.00%).
- **C2->C1**: 145/412 regions (35.19%).

Common-weight BoD decomposition

The annual BoD family scores are constructed with one common weight vector per family. Script 07.1 preserves the underlying $\text{DMU} \times \text{year} \times \text{variable} \times \text{common-weight}$ decomposition for interpretation and audit, but does not reintroduce those variables as independent clustering dimensions.

Variable contributions to convergence

The table below reports the variable-level aggregation already produced by Script 06 for regions with positive overall convergence in each directional flow. Positive consistency means the variable contributed in the direction of the alternative profile for that share of converging regions.

Flow	Variable	Median contribution	Mean contribution	Consistency
C1->C2	poor420	0.091587	0.229204	81.08%
C1->C2	gini	0.055529	0.066274	94.59%
C1->C2	prosgap2021	0.012206	0.026530	89.19%
C1->C2	share_15_64	0.000021	0.003424	51.35%
C1->C2	ntl_per_capita_norm	-0.000189	-0.000668	48.65%
C1->C2	share_65_plus	-0.003139	-0.003018	29.73%
C2->C1	poor420	0.030554	0.085677	96.55%
C2->C1	share_15_64	-0.000818	0.003291	48.97%
C2->C1	share_65_plus	-0.001027	-0.000690	34.48%
C2->C1	ntl_per_capita_norm	-0.001223	-0.000610	26.90%
C2->C1	prosgap2021	-0.001865	0.002429	34.48%
C2->C1	gini	-0.009436	-0.012271	20.69%

Country-reference and local-model context

The region $\times$ country-reference analysis contains **14,336 comparisons**. The leave-one-out own-country reference is the nearest reference for **188/512 regions**.

Independent intracountry M-Exp-FCMd models are technically viable for **18/28 countries**. Global-vs-local ARI is descriptive only, especially where the locally selected number of clusters differs from the global C=2 solution.

Interpretation constraints

- Clusters are descriptive trajectory profiles, not an overall ranking of regions.
- C1->C2 and C2->C1 denote relative convergence toward a fixed alternative medoid profile, not cluster switching.
- Common-weight BoD contributions explain family-score variation; they are not extra M-Exp-FCMd dimensions.
- Structural variables are retained as contextual descriptors only and do not affect the frozen global clustering solution.

Consolidated artifacts

- Workbook: `outputs/07.1_consolidated_results.xlsx`
- Machine-readable tables: `data/interim/script07/`